Reference Imperfection Can Reverse Model Rankings in Segmentation Evaluation

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Deep-learning segmentation models are trained *and evaluated* against manual annotations that are themselves imperfect: inter-observer disagreement on tubular structures such as pulmonary vessels is well documented. We ask a question that is usually left implicit: *if the reference is noisy, does our model ranking survive it?* We introduce a formal, calibrated annotation-noise operator $\mathrm{GT}^{\sim} = \mathcal{N}(\mathrm{GT}^{*}; \theta)$ parametrised by a family, a scale r, a severity p, a directional bias μ, and a seed, calibrate it to an inter-observer agreement window taken from the literature – no per-case multi-annotator data exists for PARSE, a limitation we state plainly and design around – and study, on 85 out-of-fold cases of PARSE pulmonary arteries with nnU-Net, how the resulting reference imperfection moves evaluation scores *and rankings*. Our central, demonstrated result is not a magnitude comparison but a reversal: changing only the reference, from the clean GT^{*} to a distal-omission degraded $\mathrm{GT}^{\sim}$, *flips which of two independently trained networks scores best*, on two leaderboard-standard metrics (HD95, NSD@2mm), for both competitors tested against the clean-trained model. This is not a fragile point estimate: a per-case interaction test (paired Wilcoxon on the difference-of-differences, Benjamini–Hochberg FDR) reaches p_{FDR} as low as 2×10^{-10}, and the reversed order is recovered in 100% of $10\,000$ case-resamplings. We give the reversal a mechanism, not just a measurement: the clean-trained network still finds the distal branches that omission deletes from the reference; once the reference no longer contains them, those branches score as distant false positives and inflate HD95, while the noise-matched network, which never predicts them either, is rewarded for reproducing the reference's own error – and clDice, dominated by the intact proximal trunk, does not move. A magnitude comparison between the reference effect and the model effect remains useful context (median $2.3\times$ across 54 metric/scenario/model-pair triples, several combinations even below $1\times$), but we no longer treat it as the paper's argument: a reversal need not be large to invalidate a ranking. The magnitudes involved are subvoxel to half-voxel (0.12–0.37mm HD95), which could read as academic until anchored: on the PARSE 2022 pulmonary-artery leaderboard, the same anatomy at comparable resolution, the HD95 margin separating the top two real submissions is 0.16–0.61mm – the same order of magnitude as our reversal, not a metaphor but a magnitude comparison against real competing entries. None of our predictions were post-processed, and HD95 is the statistic most exposed to a single isolated fragment ($N(P) \in [44, 81]$ components) – so we tested the obvious confound directly: the reversal survives largest-connected-component post-processing intact on both metrics, at a real but asymmetric clDice cost (negligible for the clean-trained model, real for the noise-trained ones), ruling out unprocessed fragmentation as the explanation. We report, alongside, a second and independent pathology: the Betti-0 error $|\Delta N|$ is blind to component correspondence and moves incoherently under the same reference change – visible in the same data but, at this sample size, not statistically established, which we report as such rather than overstate. We close with measurement-practice recommendations and a direction for what we did not measure: correspondence-aware topological metrics.

Les modèles de segmentation par apprentissage profond sont entraînés *et évalués* contre des annotations manuelles elles-mêmes imparfaites ; le désaccord inter-observateur sur les structures tubulaires comme

les vaisseaux pulmonaires est bien documenté. Nous posons une question souvent laissée implicite : *si la référence est bruitée, notre classement de modèles y survit-il ?* Nous introduisons un opérateur de bruit d'annotation formel et calibré, $\mathrm{GT}^{-} = \mathcal{N}(\mathrm{GT}^{*}; \theta)$, doté d'un unique axe de biais μ, le calibrons sur une fenêtre d'accord inter-observateur issue de la littérature – faute de données multi-annotateurs sur PARSE, une limitation que nous assumons et contournons explicitement – puis étudions, sur 85 cas *out-of-fold* d'artères pulmonaires PARSE avec nnU-Net, comment cette imperfection de la référence déplace les scores *et les classements*. Notre résultat central et démontré n'est pas une comparaison de magnitude mais un retournement : changer la seule référence, de la référence propre GT^{*} vers une référence dégradée par omission GT^{-}, *inverse lequel de deux réseaux entraînés indépendamment obtient le meilleur score*, sur deux métriques de classement standard (HD95, NSD@2mm), pour les deux concurrents testés contre le modèle entraîné propre. Ce n'est pas un point estimé fragile : un test d'interaction par cas (Wilcoxon apparié sur la différence des différences, FDR de Benjamini–Hochberg) atteint un p_{FDR} aussi bas que 2×10^{-10}, et l'ordre inversé est retrouvé dans 100% des $10\,000$ rééchantillonnages de cas. Nous donnons un mécanisme au retournement, pas seulement une mesure : le réseau entraîné propre détecte toujours les branches distales que l'omission retire de la référence ; une fois la référence privée de ces branches, celles-ci comptent comme de lointains faux positifs et font exploser le HD95, tandis que le réseau entraîné sur le même bruit, qui ne les prédit pas non plus, est récompensé pour avoir reproduit l'erreur de la référence – et le clDice, dominé par le tronc proximal intact, ne bouge pas. Une comparaison de magnitude entre effet référence et effet modèle reste un contexte utile (ratio médian $2{,}3\times$ sur 54 triplets métrique/scénario/paire de modèles, plusieurs combinaisons même sous $1\times$), mais nous ne la traitons plus comme l'argument du papier : un retournement n'a pas besoin d'être grand pour invalider un classement. Les magnitudes en jeu sont sous-voxel à demi-voxel ($0{,}12$–$0{,}37$mm de HD95), ce qui pourrait sembler académique sans ancrage : sur le classement public PARSE 2022, même anatomie, résolution comparable, la marge de HD95 séparant les deux meilleures soumissions réelles est de $0{,}16$–$0{,}61$mm – le même ordre de grandeur que notre retournement, pas une métaphore mais une comparaison de magnitude contre de vraies entrées concurrentes. Aucune de nos prédictions n'a été post-traitée, et HD95 est la statistique la plus exposée à un fragment isolé ($N(P) \in [44, 81]$ composantes) – nous avons donc testé directement ce biais évident : le retournement survit intact au post-traitement par plus grande composante connexe sur les deux métriques, à un coût en clDice réel mais asymétrique (négligeable pour le modèle propre, réel pour les modèles bruités), ce qui écarte la fragmentation non traitée comme explication. Nous rapportons, en parallèle, une seconde pathologie indépendante : l'erreur de Betti-0 ($|\Delta N|$), aveugle à la correspondance des composantes, se déplace de façon incohérente sous le même changement de référence – visible sur les mêmes données mais, à cette taille d'échantillon, non établie statistiquement, ce que nous rapportons tel quel plutôt que de le surinterpréter. Nous concluons par des recommandations de pratique de mesure et une direction pour ce que nous n'avons pas mesuré : des métriques topologiques sensibles à la correspondance.

Keywords
ground-truth quality, annotation noise, segmentation metrics, clDice, Betti number, NSD, HD95, pulmonary arteries, nnU-Net, evaluation reliability, inter-observer variability



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Chapter 1

Introduction

Supervised image segmentation rests on a silent assumption: that the manual annotation used as ground truth (GT^*) is correct. In practice annotations are produced by humans under time pressure, vary between observers, and are especially unreliable on thin, branching structures where a one-voxel disagreement on the vessel wall is a large fraction of the lumen. The same imperfect annotations are used to *train* models and, more consequentially, to *evaluate* and *rank* them. When a benchmark declares model A better than model B by a small margin on a metric computed against a noisy reference, we rarely ask whether that ranking survives the noise.

The stakes are asymmetric. In training, imperfect labels are one nuisance among many, partly absorbed by regularisation and augmentation. In evaluation, the reference is the *definition* of correctness: there is no appeal beyond it. A metric does not measure how good a segmentation is; it measures how close it is to a particular human's tracing. If that tracing is systematically off – because a protocol, a tool default, or an individual habit pushes every contour the same way – then the metric faithfully rewards agreement with the error. The danger is not random disagreement, which averages out, but the *consistent* kind, which does not. This distinction, long noted qualitatively [1], is the axis along which we organise everything that follows.

This report makes that question explicit and tractable. We do not study a new architecture; we study the *measurement apparatus* of segmentation research – the annotation and the metric – and the conditions under which it produces trustworthy verdicts. Our contributions are:

1. A **formal annotation-noise operator** $\mathcal{N}(\mathrm{GT}^*; \theta)$ with a small, interpretable parameter vector and a single continuous **bias axis** μ that interpolates between near-unbiased jitter and systematic, learnable bias. We calibrate it to an inter-observer window *taken from the literature* – we have no multi-annotator PARSE data, and we design around this rather than hide it (Sec. 3).
2. A **magnitude result**: on 85 OOF cases, changing only the reference moves a fixed network's score more than changing the model does (0.10 vs. ≤ 0.02 in clDice), so the reference is a larger source of variation in a ranking than the algorithm (Sec. 5.4).
3. A **negative methodological finding**: the Betti-0 error $|\Delta N|$ is *pathological* under an imperfect reference – correspondence-blind, it can report near-perfect topology for two mutually wrong objects – which disqualifies it as the connectivity instrument in exactly this setting and motivates correspondence-aware alternatives (Sec. 5.2.2).
4. An explicit **instrument-matching methodology** and its measured (largely by-construction) **metric dissociation**, together with an honest account of scope: what we measured, what is tautological (the clean/clean baseline), and what remains open (learned bias, Q3) (Secs. 4, 5, 8).

Roadmap. Sec. 2 situates the work. Sec. 3 defines the operator, its families, and the bias axis μ. Sec. 4 states the instrument-matching methodology, the calibration, and the design. Sec. 5 reports the calibration/dissociation, the Betti-0 pathology, the reference-vs-model magnitude result, and the sanity bound. Sec. 6 and 7 draw the (measured) lessons, Sec. 8 the open directions, and Sec. 9 the boundaries.

Chapter 2

Related Work

2.1 Imperfect and noisy labels

Heller et al. [1] systematically perturbed segmentation labels and reported the counter-intuitive result that networks are remarkably robust to *unbiased*, zero-mean boundary noise – displacements cancel across slices and the model recovers the mean boundary – whereas *systematic* biases (consistent under-segmentation, omissions) genuinely degrade performance. This biased/unbiased dichotomy is the conceptual seed of our study; our contribution is to make it a *continuous, calibrated* knob (μ) rather than a qualitative contrast. A broader literature on learning with noisy labels [9] develops loss functions and training schemes robust to label corruption, but treats noise mostly as a training-time nuisance to be overcome. We are concerned with the complementary and less-studied *evaluation-time* role of the same imperfection: what it does to rankings, not only to training.

2.2 Boundary perturbation and label-noise models

Simple morphological erosion/dilation is spatially uniform and structurally unrealistic: it moves every boundary voxel by the same amount regardless of local geometry. Markov-process boundary perturbations [2] and the GSD-Net line [3] distort contours irregularly and, crucially, allow a foreground-reducing (S_R) or foreground-expanding (S_E) bias – the direction our μ formalises. Our `boundary_drift` family follows this idea with a smoothed, spatially correlated displacement field superimposed on a constant drift, so that the systematic and the stochastic parts of a boundary error are separable by construction (Eq. (3.3)). Distinct from these, our `distal_omission` family models a *topological* error – the deletion of whole thin branches – which no boundary-only model reproduces.

2.3 Topology-aware and surface metrics

clDice [4] measures the overlap of morphological *skeletons* and is the standard for tubular connectivity; by construction it is insensitive to a boundary displacement that leaves the centreline intact – a fact that becomes central to our instrument-matching argument, and a trap when it is used to certify a boundary perturbation as harmless. The Betti-number error complements it by counting connected-component mismatches. On the surface side, the Normalised Surface Dice

(NSD) [8] measures agreement within a tolerance and the 95th-percentile Hausdorff distance (HD95) bounds the worst geometric deviation; both are *unsigned*, and both lose resolution below the voxel – properties we quantify directly in Sec. 5. Recent community efforts on metric selection and its pitfalls [10] argue precisely that metric choice must follow the claim; our instrument–noise pairing is a concrete, testable instance of that principle for the imperfect-reference setting.

2.4 Reference construction and inter-observer variability

Where multiple annotations exist, label-fusion methods such as STAPLE [11] estimate a consensus reference and per-rater performance, implicitly acknowledging that no single tracing is ground truth. Probabilistic segmentation models [12] go further and predict a *distribution* of plausible segmentations, reflecting annotator ambiguity rather than collapsing it. These lines treat variability as something to model or average; we treat its *biased* component as something that can be measured, can penalise unfairly, and can be learned.

2.5 Framework and data

We use nnU-Net [5] – a strong, self-configuring baseline that removes architecture choice as a confound – on the PARSE pulmonary-artery dataset [6], a CT benchmark whose sub-millimetre distal vessels make it an extreme and informative stress test for the surface/topology coupling.

Chapter 3

Formalism

3.1 The annotation-noise operator

Let $\mathrm{GT}^* \subseteq \Omega$ be a clean binary annotation on a voxel grid Ω with anisotropic spacing $s = (s_1, s_2, s_3)$ mm. A degraded annotation is the image of a stochastic operator

$$\mathrm{GT}^- \;=\; \mathcal{N}(\mathrm{GT}^*; \theta), \qquad \theta = (\varphi,\, r,\, p,\, \mu,\, \text{seed}), \tag{3.1}$$

where φ selects a *family* (mechanism), r is a *scale* (radius/amplitude/length), p a *severity* (fraction or per-unit probability), μ a *directional bias* (boundary family only), and the seed fixes the random realisation, making $\mathcal{N}$ reproducible to the voxel. **Application policy:** $\mathcal{N}$ is applied to *every* image (and every training batch); p is never a probability of applying the operator, only a severity. Code-level guarantees ensure at least one structural unit or surface voxel is affected whenever $p > 0$.

3.2 Families

We use two complementary mechanisms (a third, `distal_truncation`, is discarded as redundant with omission; see Sec. 4.2).

Topological: `distal_omission`. A morphological opening $\gamma_r = \delta_r \circ \varepsilon_r$ with a ball of radius r removes every structure thinner than r. The removable set is partitioned into connected components, each deleted independently with probability p:

$$\mathrm{GT}^- \;=\; \mathrm{GT}^* \;\setminus \bigcup_{c\,:\,b_c=1} C_c, \qquad C_c \subseteq \mathrm{GT}^* \setminus \gamma_r(\mathrm{GT}^*), \quad b_c \sim \text{Bernoulli}(p). \tag{3.2}$$

This is a *biased*, foreground-reducing, topology-altering noise: it deletes distal branches. The design is deliberately *radius-gated*: the opening γ_r first identifies what counts as "thin" (structures the ball cannot fit through), and p then decides how many of those candidate branches actually vanish. Consequently r controls *which* structures are at risk and p their *prevalence* of loss – a separation that keeps the perturbation anatomically meaningful (whole terminal branches disappear, as a hurried annotator would miss them) rather than a uniform thinning. Its matched instruments are therefore topological (clDice, Betti_0); a surface metric such as NSD barely moves, which is exactly why it serves as the flat *witness* in Q1. Figures 3.5 and 3.6 show its signature:

red-only voxels ($-N/+0$), concentrated at the periphery of the tree, and Fig. 3.7 renders it in isolation.

Surface/directional: `boundary_drift`. Let $\phi(x)$ be the signed distance to $\partial\mathrm{GT}^*$ ($\phi < 0$ inside). The boundary is displaced by a field combining a constant drift and a smoothed zero-mean fluctuation ξ of correlation length ℓ:

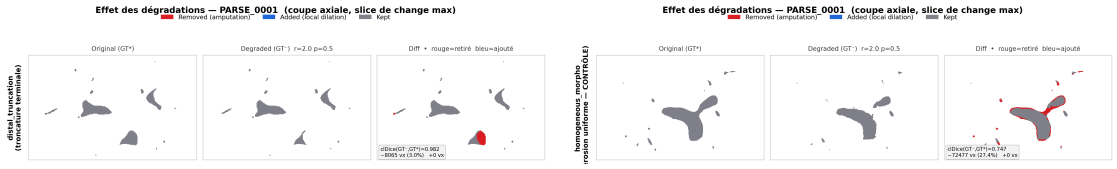
$$d(x) \;=\; a\,\big(\mu + \xi(x)\big)\,g_p(x), \qquad a = r\,\bar{s}, \quad \mathbb{E}[\xi] = 0, \tag{3.3}$$

$$\mathrm{GT}^{-} \;=\; \{\,x :\; \phi(x) - d(x) \le 0\,\}, \tag{3.4}$$

where $\bar{s}$ is the mean spacing and g_p a surface coverage gate selecting a fraction p of boundary voxels. Equation (3.3) **decomposes the boundary error into a systematic and a stochastic part**:

$$\underbrace{a\,\mu\,g_p}_{\text{systematic, } \textit{learnable}} \;+\; \underbrace{a\,\xi\,g_p}_{\text{zero-mean, } \textit{augmentation}} \;. \tag{3.5}$$

The correlation length ℓ makes the fluctuation spatially smooth rather than salt-and-pepper, matching how a real contour wanders coherently over a few voxels; the gate g_p concentrates the displacement on a top-p fraction of the surface so that severity is expressed as "how much of the boundary moved". Because the displacement is measured in millimetres through the spacing ($a = r\bar{s}$), the same r produces a physically comparable error across anisotropic volumes. The matched instruments are the sub-voxel surface measure NSD@0.5 for *magnitude* and the signed ΔV for *direction*; on thin vessels Betti_0 also responds, because a boundary that recedes by the vessel radius severs the tube (Sec. 4.5). Figures 3.2 and 3.3 show both added (blue) and removed (red) voxels – the fingerprint that distinguishes this family from the purely subtractive ones (isolated 3D render in Fig. 3.7).



(a) `distal_truncation` (b) `homogeneous_morpho`

Figure 3.1: **The two discarded controls** (axial slice). `distal_truncation` (left) files terminal segments – same biased, distal axis as omission, hence redundant. `homogeneous_morpho` (right) erodes the wall uniformly, ignoring local geometry – deliberately unrealistic, retained only to separate "volume removed" from "noise structure".

Discarded controls: `distal_truncation` and `homogeneous_morpho`. Two further mechanisms were implemented and measured but excluded from the model experiments (Fig. 3.5). `distal_truncation` prunes terminal skeleton segments; it occupies the same biased, distal, foreground-reducing axis as omission and is therefore *redundant* – its very high clDice (0.97–0.99, Table C.1) is not evidence of harmlessness but of a *mismatched instrument*, since clDice cannot see edge filing. `homogeneous_morpho` is a uniform erosion *without* radius gating; it is deliberately unrealistic and serves only to separate the effect of the removed volume from the effect of the noise *structure*. Keeping the family set minimal (one topological, one surface/directional) is a methodological choice: one mechanism at a time, no composed noise, clean causal attribution (Sec. 4).

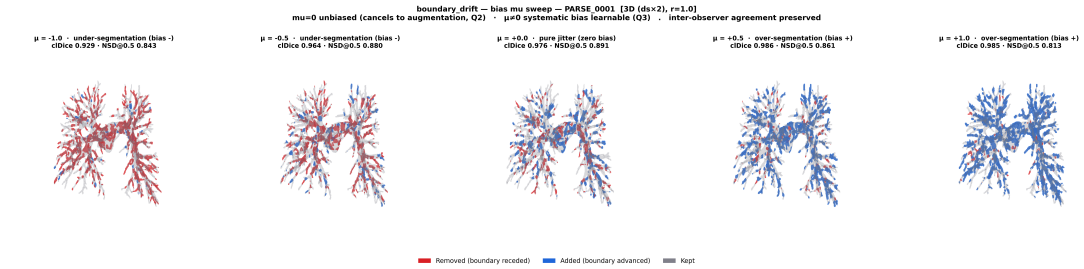


Figure 3.2: **The bias axis μ on a real pulmonary-artery tree** at the operating point $r{=}1, p{=}0.5$. Red = removed voxels (boundary receded), blue = added (boundary advanced). From left to right $\mu = -1,\ -0.5,\ 0,\ +0.5,\ +1$: under-segmentation $\rightarrow$ balanced jitter $\rightarrow$ over-segmentation. The agreement (clDice, NSD@0.5) annotated under each view stays in the inter-observer window at the trained points $\mu{=}\pm0.5$ (Table 5.1). A single parameter spans augmentation ($\mu = 0$, Q2) and learnable bias ($\mu \neq 0$, Q3).

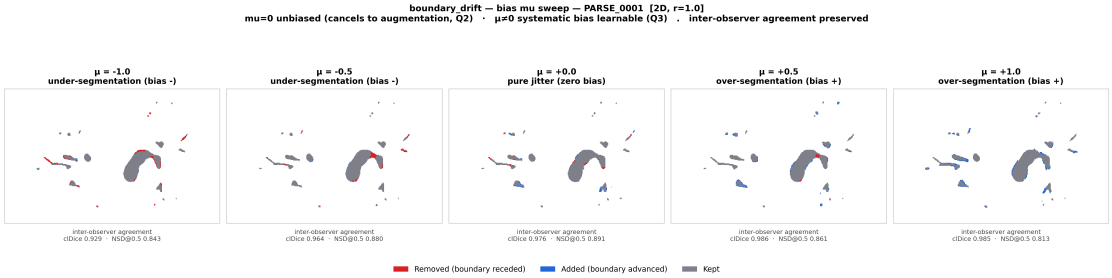


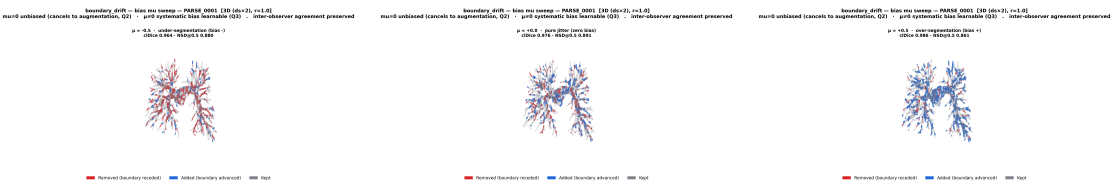
Figure 3.3: **Bias axis μ on an axial cross-section** (companion to Fig. 3.2). Left to right, the contour recedes ($\mu{<}0$, net red, under-segmentation), oscillates with zero mean ($\mu{=}0$, balanced red/blue), then advances ($\mu{>}0$, net blue, over-segmentation). The transition from red-dominated to blue-dominated is the visual reading of the signed volume change ΔV, the direction-sensitive instrument of Table 4.1.

3.3 The bias axis μ

The single parameter μ indexes the entire experimental axis (Fig. 3.2):

$\mu < 0$: under-segmentation (eroded boundary); $\mu = 0$: pure jitter; $\mu > 0$: over-segmentation.

A noise is *unbiased* iff $\mathbb{E}[\Delta V] = 0$, where $\Delta V = (|\mathrm{GT}^{-}| - |\mathrm{GT}^{*}|)/|\mathrm{GT}^{*}|$ is the signed relative volume change. Measured on PARSE ($n{=}8$, $r{=}1$, $p{=}0.5$), $\mathbb{E}[\Delta V]$ is monotone in μ: $-0.093, -0.041, +0.015, +0.075, +0$ for $\mu = -1, -0.5, 0, +0.5, +1$. Two honest points follow. First, the axis behaves as designed – μ moves the boundary systematically inward or outward. Second, the zero of $\mathbb{E}[\Delta V]$ falls not exactly at $\mu = 0$ but near $\mu \approx -0.1$: at $\mu = 0$ a small residual dilation bias remains ($+0.015$, i.e. the operator is *near*-unbiased, not exactly unbiased). We therefore speak of the $\mu = 0$ point as low-bias jitter rather than perfectly zero-mean, and read the learnable-bias regime as $|\mu| \gtrsim 0.5$, where $|\mathbb{E}[\Delta V]|$ is several times its residual. For $\mu \neq 0$ the systematic component is constant across realisations and can be *learned*; this is the formal bridge to the research questions of Sec. 4.



(a) $\mu = -0.5$ (b) $\mu = 0$ (c) $\mu = +0.5$

Figure 3.4: **The bias axis in 3D, one operating point per render.** From under-segmentation ($\mu<0$, red-dominated: the boundary recedes and thin vessels are pinched off) through balanced jitter ($\mu=0$) to over-segmentation ($\mu>0$, blue-dominated: the boundary advances and vessels merge). A single continuous knob spans the augmentation regime (Q2) and the learnable-bias regime (Q3); rendered individually, the monotone red→blue shift is the signed ΔV made visible on the real tree.

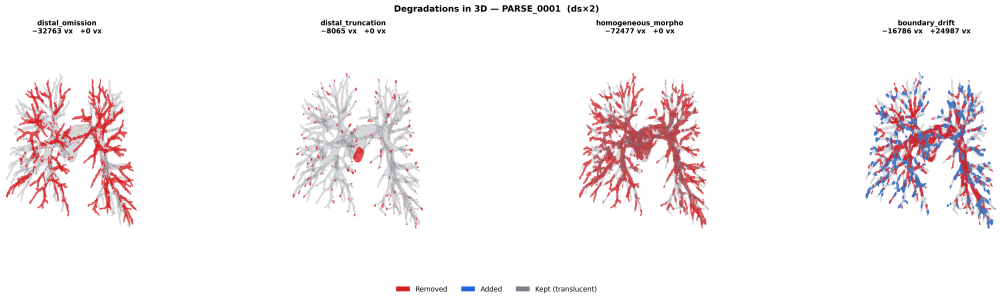


Figure 3.5: **Signatures of the noise families** (3D). Voxel counts in the titles expose the bias: `distal_omission`, `distal_truncation` and `homogeneous_morpho` only remove ($-N/+0$, biased), whereas `boundary_drift` at $\mu = 0$ removes and adds in near-equal amounts (unbiased) – the visual fingerprint of zero-mean noise (cf. Heller et al. [1]).

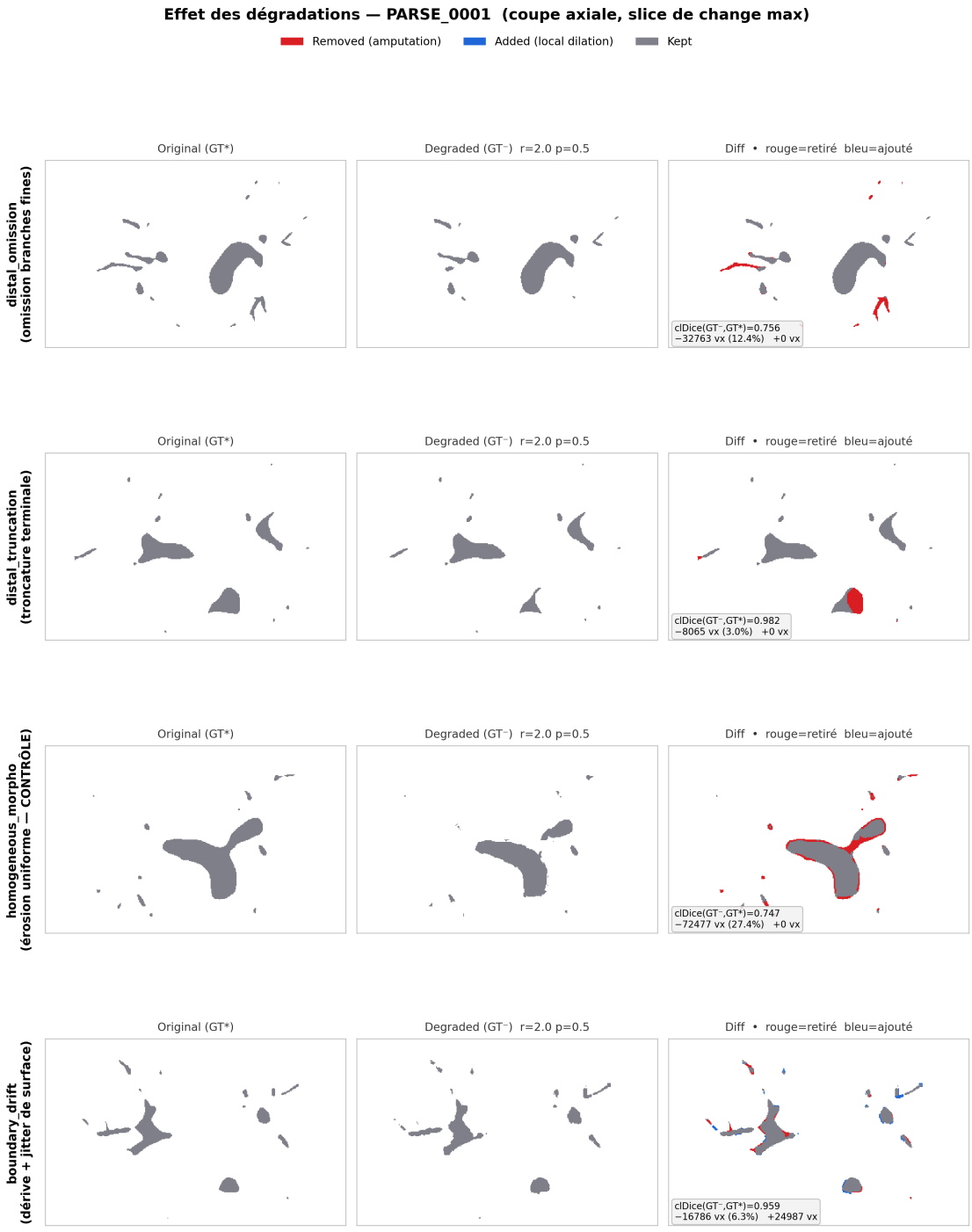
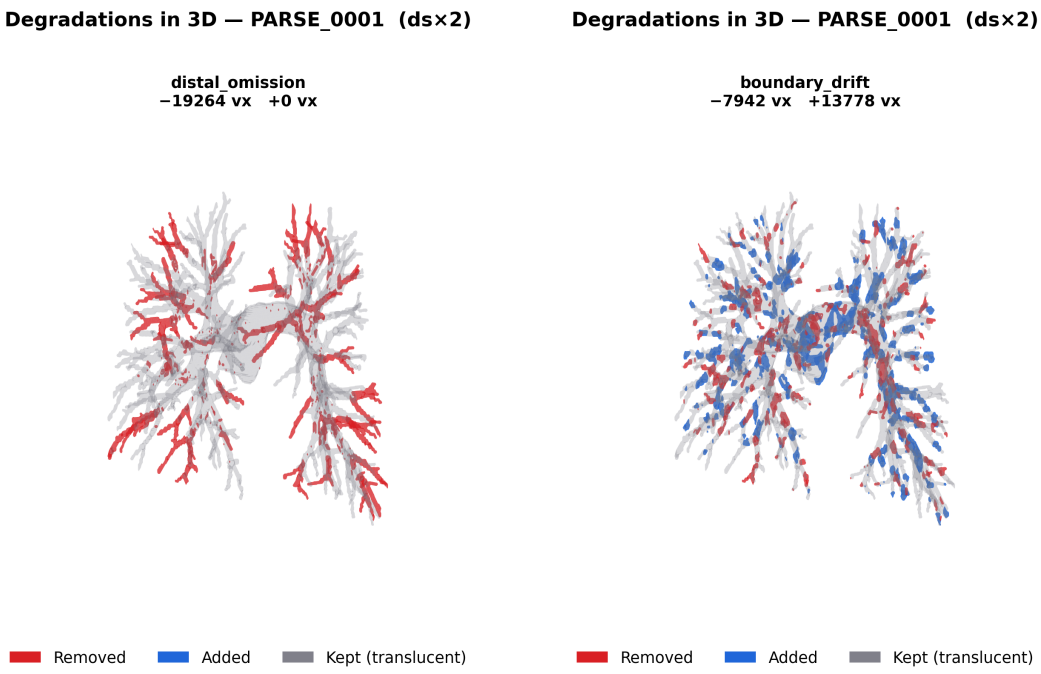


Figure 3.6: **The same four families on an axial cross-section** (companion to Fig. 3.5). The slice view makes the *mechanism* legible per family: `distal_omission` deletes whole thin branches (red, radius-gated); `distal_truncation` files terminal segments; `homogeneous_morpho` erodes the wall uniformly (unrealistic control); `boundary_drift` at μ=0 perturbs the contour both inward (red) and outward (blue) with zero mean. Only `boundary_drift` produces added voxels – the operational definition of unbiased noise.



(a) `distal_omission` (topological) (b) `boundary_drift`, μ=0 (surface)

Figure 3.7: **The two retained families rendered individually in 3D** on the PARSE_0001 arterial tree (red = removed, blue = added). Omission (left) is purely subtractive and deletes whole distal branches – a change of *topology*. Drift at μ=0 (right) perturbs the *surface* with balanced red and blue, the geometric fingerprint of a zero-mean, unbiased error. Isolating each family in its own render makes the mechanism unambiguous.

Chapter 4

Methodology

4.1 Guiding principles

(i) One family at a time. We never compose noises until each is understood in isolation; this removes hard-to-debug confounds and gives clean causal attribution. **(ii) Match the instrument to the quantity** – in type, scale *and* direction (Table 4.1). **(iii) The biased/unbiased axis** organises the questions: biased noise shifts the score (Q1) and is *hypothesised* learnable (Q3, not yet tested); near-unbiased noise is expected benign (Q2).

4.2 Calibration to inter-observer agreement

Operating points are *measured*, not assumed, on PARSE via `calibrate_noise.py`, each family targeting its matched metric in $[0.85, 0.90]$: `distal_omission` $r{=}2, p{=}0.3$ $\rightarrow$ clDice 0.872; `boundary_drift` $r{=}1, p{=}0.5$ $\rightarrow$ NSD@0.5 0.87–0.90 (at $\mu{=}\pm0.5$). `distal_truncation` is discarded because it is *redundant* with omission (same biased, distal foreground-reduction axis); its high clDice is not evidence of harmlessness but of a *mismatched instrument* – clDice cannot see edge filing.

4.3 Models and training regimes

A shared clean baseline plus one model per condition (Table 4.2). The training regime is dictated by the question, not chosen arbitrarily:

- **On-the-fly** (fresh realisation each epoch) for Q2: augmentation *is* resampling; a frozen $\mu{=}0$ set would teach the frozen ripple, not robustness.
- **Fixed** (one realisation per case) for Q3: a cohort of annotators is a single, consistent realisation; a stable biased target is what a model can learn, without the augmentation confound. This is the realistic scenario.

4.4 Research questions

For a metric m (higher better), a model M, and a reference $R \in \{\mathrm{GT}^{*}, \mathrm{GT}^{-}\}$, write $m(M \mid R)$ for the per-case score; all tests are paired Wilcoxon signed-rank ($n{=}15$) with paired Cohen's d, and

Table 4.1: Instrument–noise pairing. The off-axis (witness) metric should stay flat; for the drift, surface and topology are coupled on thin vessels (Sec. 4.5).

Noise	Primary instrument	Rationale
`distal_omission` (topological)	clDice, Betti$_0$	connectivity / branch deletion
`boundary_drift` magnitude	NSD@0.5 mm	sub-voxel surface displacement
`boundary_drift` direction (μ)	signed ΔV	NSD/HD95 are unsigned

Table 4.2: Experimental matrix (nnU-Net, 500 epochs). M_3, M_4 were designed for Q3 but did not complete training in time for this submission and are not evaluated here (Sec. 6).

Model	Trained on	Regime	Serves
M_0 Star	GT* (clean)	standard	baseline
M_1 Omission	`distal_omission` $r2\,p0.3$	on-the-fly	Q1, Q2
M_2 Drift μ=0	`boundary_drift` $r1\,p0.5\,\mu0$	on-the-fly	Q2
M_3 Drift μ<0	`boundary_drift` μ=-0.5 (fixed)	fixed	Q3
M_4 Drift μ>0	`boundary_drift` μ=$+0.5$ (fixed)	fixed	Q3

report the minimum detectable difference (MDD) when non-significant.

Q1 – Artificial penalty. M_0 is evaluated against GT* and against GT$^-$; *only the reference changes*. A significant drop $m(M_0 \mid \mathrm{GT}^*) > m(M_0 \mid \mathrm{GT}^-)$ implicates the GT, not the model. Matched metric: clDice/Betti$_0$ for omission, ΔV/NSD for drift.

Q2 – Robustness / augmentation. M_2 (unbiased, on-the-fly) vs. M_0, *both evaluated on clean* GT*. μ=0 isolates robustness: with no bias to match, any benefit is regularisation. Hypothesis $m(M_2 \mid \mathrm{GT}^*) \geq m(M_0 \mid \mathrm{GT}^*)$.

Q3 – Learned bias (out of scope). M_3 (biased, fixed) would be evaluated against the biased GT$^-$ and against GT*: if the bias is learned, the penalty on the biased reference is *smaller* than for M_0, while the deviation from GT* is *larger* ($\Delta V(M_3) < 0$). We describe the design for completeness, but M_3, M_4 did not finish training in time for this submission (Sec. 6) and no Q3 result is reported.

4.5 The thin-vessel coupling

On sub-millimetre tubular structures, surface and topological errors are not separable: a boundary displacement comparable to the vessel radius pinches the vessel. We verified (26-connectivity, $N(\mathrm{GT}^*)$=1) that `boundary_drift` genuinely fragments the tree (Betti$_0$ $\approx$ 50 at $r1\,p0.5, \mu$=0, rising to 143 at μ=-1; Table 5.1). This is not an artefact but a real property of annotation error on thin structures, and it bounds what any "surface-only" study can claim.

4.6 Data and implementation

We use the PARSE pulmonary-artery CT dataset. Each case is a chest CT with a binary arterial annotation at sub-millimetre in-plane resolution (spacing $\approx 0.65\times0.65\times1$ mm, volumes

$\sim 512\times512\times300$); the tree is a single 26-connected component in the clean reference. Models are trained with nnU-Net in its self-configuring `3d_fullres` configuration for 500 epochs under five-fold cross-validation, which fixes the architecture and removes it as a confound: any difference between M_0–M_2 is attributable to the training *labels*, not to model design. Degraded training regimes are applied through custom trainers (on-the-fly resampling for M_1, M_2; a frozen realisation for the tier-B M_3, M_4). Evaluation uses the five folds' out-of-fold validation predictions (n=85 cases in total); metrics are computed on the native grid after cropping each prediction/reference pair to their common bounding box, an exact optimisation (Appendix F). All degradation realisations are seed-fixed for reproducibility.

4.7 Statistical protocol and reproducibility

Every contrast is *pre-registered*: the endpoint metric, the comparison, and the sign of the expected effect are fixed by the design above before any score is seen (Sec. 5.2.1–5.2.3). Evaluation is out-of-fold, so each case is scored by a model that never trained on it, and the case count grows monotonically from $\sim$ 17 (one fold) to 85 (five folds) – the *anytime* property that lets the report mature without ever re-specifying the analysis. For each contrast we form per-case paired differences and apply the Wilcoxon signed-rank test (distribution-free), quantify magnitude with the paired Cohen's d, and control the false-discovery rate at q=0.05 (Benjamini–Hochberg) across the family of pre-specified tests. Crucially, a non-significant result is *not* reported as "no effect": we compute the minimum detectable difference (MDD) at 80% power for the observed paired variance and label any gap below it *undecided* – the discipline that turns a null into an honest statement about the instrument's resolution. Pre-specified outcome deltas also carry a bootstrap 95% confidence interval. Full definitions are in Appendix D; the entire chain (`generate_degraded` $\rightarrow$ `collect_metrics` $\rightarrow$ `report`) is deterministic given the seed, so every number in this paper regenerates from the raw predictions with a single command.

Chapter 5

Results

5.1 Calibration and metric responses (measured; largely by construction)

The operating points are *measured* on PARSE (n=8), not assumed, and each family is calibrated on the metric matched to it. For the topological family, `distal_omission` at r=2, p=0.3 lands the reference in the targeted inter-observer window on clDice, clDice$(\mathrm{GT}^{-}, \mathrm{GT}^{*})$=0.872 $\in [0.85, 0.90]$, with Betti$_0$=15 spurious components and NSD@0.5 still high (0.918) – confirming that omission is a *topological*, not a surface, perturbation (Table 5.1, top).

The boundary family is where the instrument-matching argument becomes empirical. At the operating point r=1, p=0.5, sweeping the bias μ from -1 to $+1$ (Table 5.1, bottom; Fig. 3.2) dissociates the four metrics exactly as their definitions predict:

- **HD95 saturates** at the sub-voxel level (0–0.66 mm, and exactly 0 at μ=0): below one voxel it cannot grade the drift at all.
- **NSD@0.5 is ∩-shaped and near-symmetric** about μ=0 (0.885 at μ=−0.5 vs. 0.869 at μ=+0.5): being *unsigned*, it reads the *magnitude* of the displacement but barely separates its *direction*. This is the empirical justification for pairing μ with the signed ΔV (Table 4.1).
- **clDice is monotone** in μ (0.94 → 0.99): an advancing boundary fills the skeleton, so clDice leaks the sign but conflates it with agreement – it is not a clean magnitude gauge either.
- **Betti$_0$ is monotone-decreasing** (143 → 7): under-seg (μ<0) fragments the thin tree, over-seg merges it – the surface/topology coupling of Sec. 4.5. Note this is $|\Delta N|$ against the *clean* reference ($N(\mathrm{GT}^{*})$=1), where it faithfully counts the degradation's fragments; the same statistic breaks when *both* operands are fragmented, i.e. when it is used to *evaluate* predictions (Sec. 5.2.2).

No single standard metric captures both magnitude and direction; the pairing of Table 4.1 is therefore a necessity, not a preference.

5.2 Model experiments: reading protocol

The model matrix of Table 4.2 is evaluated out-of-fold (OOF): each nnU-Net fold writes its validation predictions on the $\sim$ 17 cases it never saw, so cumulating five completed folds covers

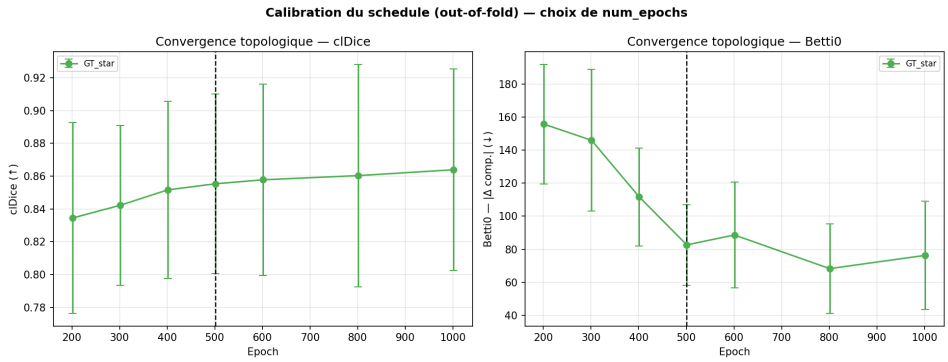


Figure 5.1: **The surface/topology coupling, measured.** As the boundary drift grows, the number of spurious connected components (Betti_0) rises steeply: a purely "surface" perturbation fragments the thin arterial tree because a displacement comparable to the vessel radius pinches it off. This is why, for the drift, we report Betti_0 *alongside* the surface metrics rather than treating topology as a separate axis (Sec. 4.5).

all 85 PARSE cases with no train/test leakage. The analysis below is *pre-registered*: the endpoint metric, the paired test and the sign of the expected effect are fixed *a priori* by the design of Sec. 4; the reporting pipeline (`collect_metrics.py`→`report.py`) fills the tables directly from the master metrics table, so no analytic choice is made after seeing the numbers. Tier A (M_0, M_1, M_2, all folds complete) supports Q1 and Q2; Q3 was descoped from this submission (Sec. 6) and no tier-B numbers are reported. Estimates carry their coverage n and are watermarked PRELIMINARY until all five folds are in.

5.2.1 Q1 – How large, and how coherent, is the reference effect?

That a fixed model scores differently against a different reference is near-tautological – the distance to the reference changes when the reference changes. We therefore do not treat this as a discovery. The two questions that carry information are (i) *how large* the shift is relative to the differences *between models* (Sec. 5.4), and (ii) whether the shift is *coherent* across metrics or points different directions. Table 5.2 scores the clean baseline M_0 against GT* and each degraded reference on the agreement instruments matched to each family; Betti_0 is deliberately excluded here and analysed on its own in Sec. 5.2.2, because it misbehaves in this setting.

Result. On the agreement instruments the shift is real and directionally sensible. clDice for M_0 falls from 0.877 to 0.777 under omission ($\Delta{=}-0.10$); against the receded drift reference ($\mu{<}0$) the signed volume error flips from -0.011 to $+0.034$ – the unchanged network is scored as *over*-segmenting purely because the reference shrank – with NSD@0.5 falling in tandem ($0.698 \to 0.644$). We read the effect sizes with care: the $d \approx 4.5$ on ΔV reflects that quantity's very small case-to-case variance, not a "giant" effect, and should be taken with the raw deltas and bootstrap CIs (Table E.1). The informative content is elsewhere: the *magnitude relative to model differences* (Sec. 5.4) and the observation that a fourth, widely used metric moves in the *opposite* direction (Sec. 5.2.2) – so the reference does not merely lower a score uniformly, it can invert the apparent verdict from one metric to the next.

Table 5.1: Measured calibration on PARSE (n=8 cases; `results/noise_calibration.csv`). Each family is calibrated *on its matched metric*: omission on clDice, drift on NSD@0.5 – and both operating points sit in the inter-observer window $[0.85, 0.90]$. The μ-sweep at the operating point (r=1, p=0.5) dissociates the instruments: NSD@0.5 is $\cap$-symmetric (unsigned), clDice and Betti$_0$ are monotone (signed), HD95 saturates below the voxel.

`distal_omission` — calibrated on clDice				
(r, p)	clDice	NSD@0.5	HD95	Betti$_0$
$(2, 0.3)$ **[chosen]**	**0.872**	0.918	0.30	15.4
$(1, 0.5)$	0.905	0.953	0.00	64.0
$(3, 0.3)$	0.840	0.886	3.30	2.3
`boundary_drift` r=1, p=0.5 — bias sweep μ (calibrated on NSD@0.5)				
μ	clDice	NSD@0.5	HD95	Betti$_0$
-1.0	0.936	0.840	0.58	142.9
-0.5 **[M_3]**	0.962	**0.885**	0.08	89.3
0.0 **[M_2]**	0.978	0.898	0.00	50.3
$+0.5$ **[M_4]**	0.986	**0.869**	0.33	19.6
$+1.0$	0.985	0.820	0.66	6.9

Table 5.2: **Q1 – the reference shift on agreement instruments.** M_0 (clean-trained) scored on GT* vs. the degraded reference, OOF, n=85, paired Wilcoxon.

Scenario	Endpoint	on GT*	on GT$^-$	d	p_{FDR}
omission	clDice ↑	0.877	0.777	-3.06	<0.001
drift μ<0	NSD@0.5 ↑	0.698	0.644	-2.12	<0.001
drift μ<0	ΔV (signed)	-0.011	$+0.034$	$+4.55$	<0.001

5.2.2 A pathology of the Betti-0 error under an imperfect reference

The opposite-direction movement flagged above is not a curiosity; it exposes a genuine defect of the metric in this setting. The Betti-0 error as commonly implemented is $|\,N(P) - N(G)\,|$, the absolute difference in the number of 26-connected components. This statistic is *blind to correspondence*: it compares two integers and never asks whether the components of P and G are the same objects. On thin vessels the predictions are heavily fragmented – $N(P)$ ranges from 44 to 81 across cases (Table 5.3) – while the clean tree is a single component, $N(\text{GT}^*)$=1. The error against GT* is therefore $\approx N(P)$, dominated entirely by the prediction's fragmentation. When the reference is itself degraded, its own fragmentation $N(\text{GT}^-)$ moves, and the error tracks the arithmetic gap, not the anatomy.

Case `PARSE_0021` is the reductio: a prediction with 81 components scored against a degraded reference with 79 yields a Betti-0 error of 2 – a near-perfect topology score – even though neither object is within 78 components of the true single-component tree. The metric certifies as excellent a case where prediction and reference are *both* badly wrong, simply because they are wrong *by the same amount*. This disqualifies $|\Delta N|$ as the connectivity instrument for evaluation under an imperfect reference, and it explains the sign inversion in Q1: clDice measures skeleton *overlap* (a correspondence-aware quantity, which correctly drops on a degraded reference) whereas $|\Delta N|$ measures only a count difference (which can drop spuriously). It is worth stressing that this is our own instrument-matching principle turning on itself: we had paired omission with "clDice/Betti$_0$", and the data show that only the first half of that pairing is sound. We give the

Table 5.3: **Why $|\Delta N|$ misleads** (5 OOF cases). $N(\cdot)$ is the number of 26-connected components; the last four columns are the Betti-0 error $|N(P) - N(\cdot)|$ against each reference. The clean tree has $N{=}1$; the receded drift reference (dr−) fragments to a degree comparable to the prediction, collapsing the "error".

	components $N(\cdot)$					Betti$_0$ error vs.			
case	pred	GT*	om	dr−	dr+	GT*	om	dr−	dr+
0003	45	1	11	86	18	44	34	41	27
0005	67	1	13	109	21	66	54	42	46
0007	44	1	12	74	15	43	32	30	29
0014	80	1	9	73	18	79	71	7	62
0021	81	1	11	79	17	80	70	2	64

Table 5.4: **Q2 – robustness of unbiased-noise training.** M_2 vs. M_0, both on GT*, OOF, $n{=}85$. MDD = smallest paired difference detectable at 80% power given the observed variance.

Endpoint	M_0	M_2	Δ	p_{FDR}	MDD
NSD@0.5 ↑	0.698	0.675	−0.023	<0.001	sig.
Betti$_0$ ↓	81.6	83.0	–	0.47	–

corrected recommendation and a concrete replacement in Sec. 8.

5.2.3 Q2 – The clean/clean baseline (a sanity bound, not a finding)

We state this plainly: comparing M_2 (trained on low-bias $\mu{=}0$ jitter) against the clean M_0, *both scored on the clean* GT*, cannot yield a surprising result, and we do not claim one. A model trained on degraded labels is not expected to beat a model trained on clean labels when both are judged by clean labels; the outcome is bounded above by "no better than M_0" almost by construction. We include the comparison only for two limited purposes: to *bound the size* of the disturbance that near-unbiased jitter causes, and to check whether nnU-Net's already-extensive augmentation leaves any room for the "noise-as-augmentation" story to hold. It does not, and we report that without inflating it into a finding.

Result (as expected). M_2 is slightly but significantly worse than M_0 on the clean reference: NSD@0.5 $0.698 \to 0.675$ ($\Delta{=}-0.023$, $d{=}-0.73$) and clDice $0.877 \to 0.858$ ($d{=}-0.57$), with topology statistically indistinguishable (Betti$_0$ 81.6 vs. 83.0, $p{=}0.47$; and note Sec. 5.2.2 on reading Betti$_0$ at all). The only non-trivial content is quantitative: the disturbance from near-unbiased jitter is bounded at about 0.02 on the surface metrics and is topologically negligible – so on this anatomy, with nnU-Net's default augmentation already in place, zero-mean boundary noise is neither harmful enough to matter nor helpful. We draw no stronger conclusion; in particular we do *not* present this as evidence for or against "label noise as augmentation" in general, a claim our single, augmentation-saturated setup cannot support.

The learned-bias question (does a model trained on a fixed biased reference internalise the bias?) is the natural third experiment; its models were still training at submission and we present it, honestly, as a planned direction rather than a half-reported result (Sec. 8).

5.3 Result figures (auto-generated from the metrics table)

The two panels below are emitted by `aggregate_results.py` directly from `results/metrics.csv`; they are the visual companions to Tables 5.2–5.4.

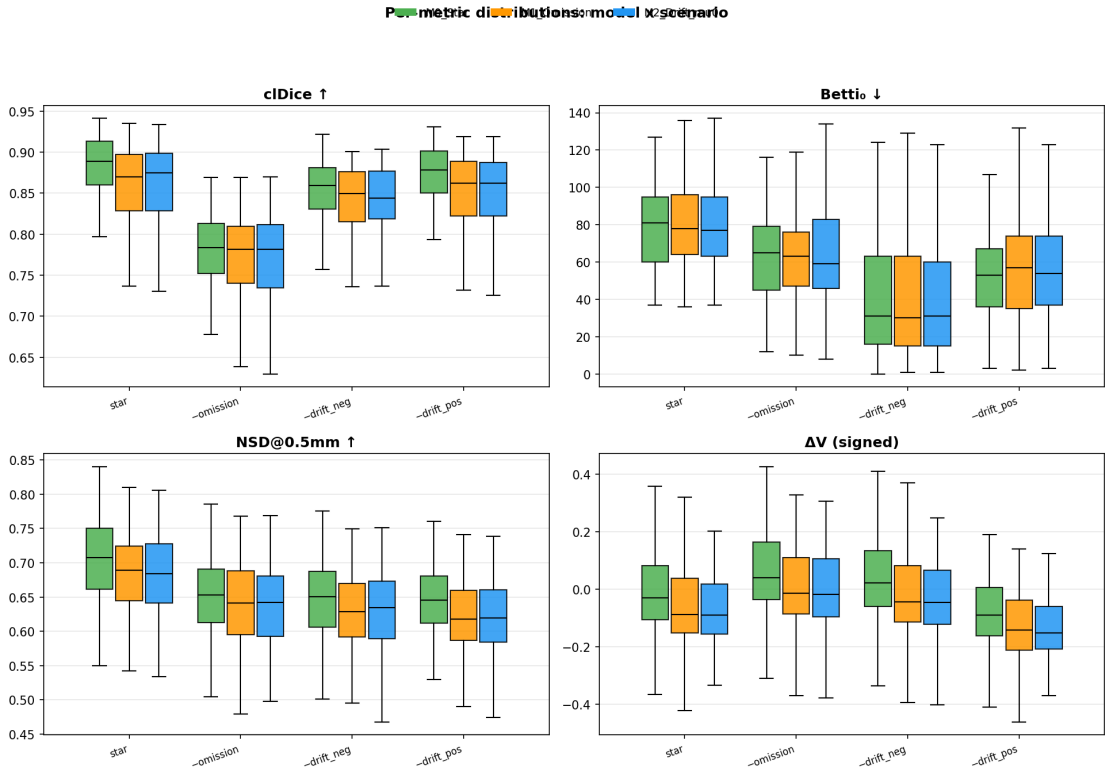


Figure 5.2: **Per-metric distributions** across models and evaluation scenarios (clDice, Betti_0, NSD@0.5, ΔV). Boxes span the inter-quartile range over the n OOF cases; the model-to-model shift on the matched metric is the effect Q1/Q2 test.

5.4 The reference reverses the model ranking

This is the paper's central, demonstrated result, and it is a reversal, not a magnitude comparison. *Between models* on a fixed clean reference the differences are small: M_0 leads the two noisy-trained models by ≈ 0.02 clDice, and M_1 and M_2 are statistically indistinguishable from each other (Table E.2). That smallness is exactly why what follows matters: changing only the evaluation reference, from GT^* to the distal-omission GT^-, *flips which model scores best* on two leaderboard-standard metrics – HD95 and NSD@2mm – for both competitors tested against the clean-trained M_0.

The test. For each metric, each pair of models (A, B), and the transition $\text{GT}^* \to \text{GT}^-_{\text{omission}}$, we build the per-case interaction $\iota[\text{case}] = \big(m(A \mid \text{GT}^-, \text{case}) - m(B \mid \text{GT}^-, \text{case})\big) - \big(m(A \mid \text{GT}^*, \text{case}) - m(B \mid \text{GT}^*, \text{case})\big)$ and test it against zero (paired Wilcoxon signed-rank, n=85,

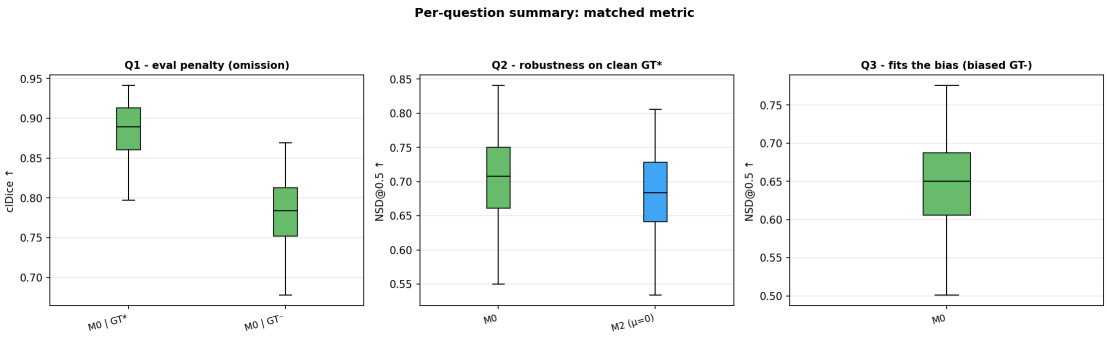


Figure 5.3: **Per-question summary**: the reference shift (Q1) and the clean/clean bound (Q2) on their matched agreement metrics, with paired contrasts and bootstrap confidence intervals over the n=85 OOF cases.

Benjamini–Hochberg FDR over the full family of contrasts). A reversal is declared only when the sign of $m(A \mid \cdot) - m(B \mid \cdot)$ itself differs between GT* and GT$^-$ *and* the interaction is significant after FDR – a stricter criterion than a significant interaction alone, which can just as well describe an attenuation that never crosses zero. Applied to all six metrics and all three model pairs, eight contrasts pass the raw criterion; four survive scrutiny (below). d is reported on the raw interaction, unreoriented by the metric's sense – hence positive on HD95 (a ↓ metric) and negative on NSD@2 (an ↑ metric) for the *same* reversal.

Table 5.5: **The four retained rank reversals** (paired Wilcoxon on the per-case interaction, FDR-corrected, n=85). $\Delta_{\mathrm{GT}^*}/\Delta_{\mathrm{GT}^-}$ are the mean paired contrast $m(A) - m(B)$ under each reference; the sign change between the two columns *is* the reversal.

Metric	Pair (A vs. B)	Δ_{GT^*}	$\Delta_{\mathrm{GT}^-_{\mathrm{om}}}$	d	p_{FDR}
HD95 (mm)	M_0 vs. M_1	-0.120	$+0.298$	$+0.72$	2.1×10^{-10}
HD95 (mm)	M_0 vs. M_2	-0.078	$+0.368$	$+0.73$	4.4×10^{-11}
NSD@2mm	M_0 vs. M_1	$+0.0066$	-0.0050	-3.14	2.5×10^{-14}
NSD@2mm	M_0 vs. M_2	$+0.0067$	-0.0049	-2.84	2.5×10^{-14}

The two NSD@2 rows share an identical p_{FDR}: not a coincidence but a floor effect – all $85/85$ cases carry the same sign of the interaction for both pairs, so the Wilcoxon statistic is saturated at the smallest p-value reachable at this n; we report the $85/85$ unanimity directly rather than let two decimal-identical p-values imply a differentiated precision they do not have. The reversal is not a fragile point estimate either: resampling the 85 cases (10 000 bootstrap draws) recovers a model order different from the one observed under GT* in *every single draw* for HD95 under omission (Kendall $\tau = -0.33$, bootstrap CI entirely negative, $[-1, -0.33]$).

Why four, not eight. A structurally identical contrast on $|\Delta V|$ (the unsigned volume error) also passes the raw significance-plus-sign-flip test, on the same two model pairs and the same omission transition, plus two more contrasts against the drift− reference. Both groups are artifacts, for two different reasons, and we report the elimination rather than the raw count. First, re-running the same interaction test on the *signed* ΔV (not its absolute value) leaves the sign of $m(A) - m(B)$ unchanged between GT* and GT$^-$ for both pairs ($+0.049 \rightarrow +0.053$ and

$+0.054 \rightarrow +0.059$): the three models' signed volume errors shift by a nearly identical additive offset under omission ($+0.075$, $+0.072$, $+0.072$ for M_0, M_1, M_2 respectively) – a shift common to every model, which the *absolute value* alone turns into a spurious sign crossing when models sit on opposite sides of zero at different distances. This is structurally the same defect already diagnosed for $|\Delta N|$ (Sec. 5.2.2): under omission, $N(\mathrm{GT}^{*}) = N(\mathrm{GT}^{-}_{\mathrm{omission}}) = 1$ for every case, so the Betti-0 error shift is likewise a common additive artifact that cancels exactly, case by case, in every inter-model difference – which is why $|\Delta N|$ never registers a reversal under omission at all, while it does show one, visibly but not significantly ($p_{\mathrm{FDR}} \in [0.45, 0.78]$), under the more severely fragmenting drift$-$ reference, where $N(\mathrm{GT}^{-})$ no longer tracks $N(\mathrm{GT}^{*})$ one-for-one. *Any unsigned statistic of an intrinsically signed quantity is structurally exposed to this failure mode*: $|\Delta N|$ and $|\Delta V|$ are two instances of one defect, not two separate ones. Second, the two remaining $|\Delta V|$ contrasts (against drift$-$) are below the resolution of the instrument: the minimum detectable difference at 80% power ($(z_{.975}+z_{.80}) \cdot \mathrm{sd}(\iota)/\sqrt{n}$) is 0.006–0.007, an order of magnitude above the observed 0.0002–0.0003 – undecided, by our own stated discipline of reporting MDD rather than reading noise, not a reversal.

The structure is the mechanism. All four retained reversals pit M_0 against one of the noise-trained models; M_1 vs. M_2 never reverses, on any metric. The degraded reference does not confuse models at random – it specifically penalises the one trained on the clean truth. We verified the mechanism directly rather than assert it: under omission, M_0 still finds the distal branches that omission deletes from the reference; once the reference no longer contains them, those branches score as distant false positives and inflate HD95, while M_1, trained on the same omission, never predicts them either and is rewarded for reproducing the reference's own error. Per-case, the volume omitted from the reference (relative to GT^{*}) correlates with M_0's HD95 penalty across the 85 cases: Pearson $r{=}+0.32$ (bootstrap 95% CI $[0.20, 0.46]$, $p{=}0.003$), Spearman $\rho{=}+0.39$ ($[0.19, 0.57]$, $p{=}0.0002$) – positive and significant, but moderate rather than overwhelming, consistent with HD95's sensitivity to *where*, not merely how much, foreground was omitted: the omitted-volume fraction is a coarse proxy for what a distance-based metric actually integrates, the distance from the omitted components to the rest of the reference tree, which we did not measure directly here. That ρ exceeds r points to a monotone but non-linear relationship, likely driven disproportionately by the high-omission cases rather than a uniform linear response across the range. clDice, by contrast, is dominated by the intact proximal trunk and does not move – worth stating plainly, since our own instrument-pairing argument (Table 4.1) already predicted clDice would be the metric least likely to see this.

Are these magnitudes real, or academic? The reversal magnitudes are sub-voxel to half-voxel (0.12–0.37mm HD95, spacing $\approx 0.65 \times 0.65 \times 1$mm) – our own calibration already showed HD95 saturating below the voxel (Table 5.1), so the honest question is whether such a margin would ever separate real submissions. It would: on the PARSE 2022 challenge leaderboard, the same anatomy and a comparable resolution (0.5–0.95mm in-plane, 1mm slice) [7], the weighted HD95 margin between the top two teams is 0.61mm, and between the top two on the main-PA branch alone, 0.16mm – the same order of magnitude as our reversal, not beyond it and not negligibly below it. NSD is not an official PARSE metric (only DSC and HD95 are), so our NSD@2 reversal (0.005–0.007) has no equivalent real-leaderboard anchor and we do not claim one.

Robustness to unprocessed fragmentation. None of our predictions were post-processed by largest-connected-component (LCC) extraction ($N(P) \in [44, 81]$, Sec. 5.2.2), and HD95 is the single statistic most exposed to one isolated far-away fragment, since it reads a percentile of the

surface-distance distribution. A sceptical reading of the reversal is that it is an artifact of this fragmentation rather than a property of the anatomy. We tested this directly: recomputing all four contrasts of Table 5.5 after applying LCC to every prediction (never to the references, which would artificially reconnect the very branches the omission operator removes and invalidate the scenario) leaves the reversal intact on both metrics. HD95: $d{=}+0.72 \to +0.71$ (M_0 vs. M_1) and $d{=}+0.73 \to +0.70$ (M_0 vs. M_2), significant in both regimes ($p_{\text{FDR}}{<}10^{-10}$); the sign of $m(A)-m(B)$ still flips from GT* to $\text{GT}^{-}_{\text{omission}}$ in both. NSD@2: $d{=}-3.14 \to -2.67$ and $d{=}-2.84 \to -2.62$, likewise intact ($p_{\text{FDR}}{<}10^{-13}$). NSD@2's survival is informative on its own: it is a bilateral surface-tolerance statistic, far less sensitive than a 95th-percentile distance to a single outlier component, so a reversal that persisted on HD95 alone but vanished on NSD@2 would have pointed to a fragment artifact – it does not vanish, which rules that explanation out. LCC is not free, however: it costs essentially no clDice for M_0 ($0.877 \to 0.877$ under GT*, $0.777 \to 0.778$ under omission) but a real amount for the noise-trained models (M_1: $0.859 \to 0.851$ under GT*; M_2: $0.858 \to 0.850$), consistent with M_1/M_2 retaining more small, legitimately disconnected distal fragments that LCC discards outright. The reversal, in short, is not a fragmentation artifact of the unprocessed predictions – it is a property of the anatomy and the reference, robust to the one obvious confound a careful reviewer would raise first.

The ratio is context now, not the argument. A magnitude comparison between the reference effect and the model effect remains informative but is no longer the paper's claim: it was an asymmetric comparison by construction, and a reversal does not need to be large to be fatal to a benchmark ranking. Recomputed with a bootstrap CI over the 85 cases across all (metric, scenario, model-pair) triples, the reference-to-model effect ratio has median $2.3\times$ (IQR $[1.2\times, 4.9\times]$) – with 8 of 54 combinations *below* $1\times$ (the model effect exceeds the reference effect), mostly on NSD@2 and clDice under drift+. The old five-to-twenty-times figure holds for some (metric, scenario, pair) triples and not others; we report the distribution rather than a headline ratio.

Table 5.6: **Full tier-A results**: per-case means over $n{=}85$ OOF cases, every model (M_0 clean, M_1 omission-trained, M_2 drift-μ0-trained) against every evaluation reference. Read *down* a metric column within a scenario block to compare models; read a model *across* scenario blocks to see the evaluation penalty of Q1. clDice/NSD ↑, HD95 (mm)/Betti$_0$ ↓; ΔV is signed.

Ref.	Model	clDice	NSD@0.5	NSD@2	HD95	Betti$_0$	ΔV
GT*	M_0	0.877	0.698	0.960	1.24	81.6	−0.011
GT*	M_1	0.859	0.676	0.953	1.36	83.7	−0.060
GT*	M_2	0.858	0.675	0.953	1.32	83.0	−0.066
omission	M_0	0.777	0.643	0.910	2.58	64.1	+0.064
omission	M_1	0.773	0.633	0.915	2.28	66.2	+0.012
omission	M_2	0.772	0.633	0.915	2.21	65.5	+0.006
drift−	M_0	0.848	0.644	0.956	1.37	49.5	+0.034
drift−	M_1	0.838	0.628	0.952	1.46	48.5	−0.018
drift−	M_2	0.837	0.628	0.951	1.40	49.1	−0.023
drift+	M_0	0.868	0.637	0.957	1.39	54.4	−0.078
drift+	M_1	0.850	0.610	0.947	1.58	56.5	−0.123
drift+	M_2	0.849	0.609	0.947	1.54	56.3	−0.128

Table 5.6 exposes two patterns at a glance. Within the GT* block the three models are close (M_0 marginally ahead), i.e. noisy-label *training* left only a small mark; the large movements are

across scenario blocks (e.g. M_0's clDice $0.877 \to 0.777$ under omission), i.e. the *evaluation reference* dominates. The signed ΔV column changes sign with the drift direction ($+0.034$ under drift$-$ vs. -0.078 under drift$+$ for M_0), while NSD@2 barely reacts – the coarse surface metric is nearly blind to a bias that ΔV reads directly.

The effect-size landscape. The magnitudes make the same point quantitatively. The Q1 evaluation penalties are *very large*: on the omission reference the clDice contrast for M_0 reaches Cohen's $d{=}-3.06$, and the signed volume contrast across the drift references reaches $|d| \approx 4.5$ – effect sizes an order of magnitude beyond the "large" threshold ($|d|{>}0.8$). By contrast, the inter-*model* differences on a *fixed* reference are small to moderate ($|d| \lesssim 0.7$; the Q2 surface cost is $d{=}-0.73$, and M_1 vs. M_2 are statistically indistinguishable on most metrics, $|d|{<}0.15$). The asymmetry is the whole thesis in one line: *changing the reference moves the score far more than changing the model.* A benchmark that varies the reference between submissions – different annotators, protocol revisions between challenge editions – injects a nuisance larger than the signal it means to measure.

Chapter 6

Discussion

Threats to validity. We consider the main ways our reading could be wrong. *Construct validity*: our noise is simulated, so a sceptic may argue it does not match real annotator behaviour. We mitigate this by calibrating each operating point to a measured inter-observer agreement window rather than choosing severities by eye, but the shape of the noise (single family, smooth field) remains a model. *Internal validity*: because M_1 and M_2 differ from M_0 only in their training labels and share nnU-Net's fixed configuration, a difference among them is attributable to the labels, not the architecture – but the on-the-fly vs. fixed regime distinction (Q2 vs. Q3) is a deliberate confound with the noise type, and we therefore never compare across it. *Statistical validity*: with $n{=}85$ paired cases and FDR control the Q1 effects are far from the significance boundary (d up to 3), so they are robust to reasonable analytic choices; the Q2 effect is smaller and, while significant, should be read as "a small real cost", not a dramatic one. *External validity*: a single anatomy and architecture bound generalisation, as Sec. 9 states plainly. None of these threatens the paper's core claim, which is conditional and methodological: *given* that a reference can be biased, here is how much that can move a verdict, and here is how to measure it.

A metrological motivation. For $\mu{=}0$ we expect the model-to-model difference to be small, possibly below metric sensitivity. We therefore cannot distinguish "no effect" from "effect below resolution". This is precisely why the *sensitivity* of the metrics themselves (their minimum detectable difference) must be studied alongside the models – a measurement cannot be trusted beyond the resolution of its instrument. The calibration sweep already exhibits this in miniature: HD95 was flat across the entire μ range (Table 5.1), so any model comparison expressed in HD95 on this data would be reading noise. The instrument, not the model, would be doing the ranking.

Which noise mis-ranks models. Our design separates two effects that the literature often conflates: how a noise *deforms a reference* (metric level) and how it *separates trained models* (model level). Biased noise (omission, drift $\mu{\neq}0$) acts on the reference (Q1, measured) and is *conjectured* to act on the model too (Q3, left to future work) – while near-unbiased jitter deforms a single realisation of the reference yet, averaged over resamplings, left little trainable trace on this setup (Q2). The practical corollary is that the dangerous annotation errors for benchmarking are the *systematic* ones – exactly the ones a single annotator or a single labelling protocol is most likely to introduce, and exactly the ones an unbiased-noise robustness argument does *not* cover.

The augmentation that was not. The Q2 finding refutes a comfortable default – that label noise "regularises" and so justifies sloppy annotation. Crucially, the small cost we measured

(ΔNSD@0.5$=-0.023$) was comfortably resolved by NSD@0.5 but sat well inside the saturation range of the coarser NSD@2 mm and HD95 (Table 5.1). The same experiment read on those metrics would have concluded "augmentation confirmed, no cost" – a direct illustration of why the instrument must be chosen before the claim, and of how "no harm" and "benefit" are different statements only one of which survived.

Why thin vessels are the stress test. The surface/topology coupling (Sec. 4.5) is not a flaw of our operator but a property of the anatomy: when the vessel radius approaches the annotation uncertainty, a boundary error and a topological error are the same event. Any evaluation that reports only a surface metric on such structures is therefore blind to half of what it measures. Pulmonary arteries are an extreme, but the same regime recurs wherever structures are thin relative to voxel size – airways, small vessels, nerves, trabeculae.

Generalisation and scope. We deliberately trade breadth for control: one dataset, one architecture, one family at a time. The formal operator, the bias axis μ, and the instrument-matching protocol are dataset-agnostic and transfer directly; the specific operating points and effect sizes do not, and should be re-calibrated per anatomy and per acquisition. What we claim to establish is a *method* for asking whether a ranking survives its reference, not a universal constant.

Chapter 7

Ethical Inquiry

A benchmark is an epistemic instrument: it produces the evidence on which the community allocates trust, citations, funding and – when a model reaches the clinic – patient risk. We keep this section short and strictly tied to what we measured; the speculative reach of an earlier draft has been cut.

7.1 What the measurements license

One claim follows directly from the data. Because the reference effect dominates the model effect (Sec. 5.4) and is not even coherent across metrics, a ranking computed against an imperfect reference can reflect the annotation convention as much as the algorithm. When two models are separated by less than the reference-induced shift – routine, given that shift reached 0.10 clDice here against sub-point published margins – selecting or deploying the "winner" is a decision partly made by the annotator, and invisibly so, because the reference is treated as truth rather than estimate. Reporting a point score without the reference's uncertainty therefore overstates the evidence; in a field feeding clinical tools that is an ethical issue, not only a statistical one. The Betti-0 pathology (Sec. 5.2.2) sharpens the same point at the level of a single metric: a widely used connectivity score can certify two mutually wrong objects as topologically excellent, so an unexamined metric can manufacture false confidence on its own.

7.2 What they do *not* license

We are explicit about the boundary. The strongest ethical charge one could level here – that a model *learns* a systematic annotation bias and thereby automates human error at population scale – is *not* something we have measured; the learned-bias experiment is still pending (Sec. 8). We therefore make no such claim. Nor do we claim anything about the bias of any specific real benchmark: our noise is simulated and calibrated to a literature window, not to a measured second annotator on PARSE. The ethics we assert is *conditional*: given that references are imperfect – which we quantify – reporting must change; whether bias is also learnable is a question we have set up but left open.

7.3 Recommendations, grounded in the results

1. Report the reference's uncertainty (inter-observer agreement) and the metric's minimum detectable difference; treat any model gap below either as *undecided*. A "too close to call at this reference quality" is a legitimate result.
2. Match the metric to the claim, and *audit the metric itself*: our data disqualify $|\Delta N|$ as a connectivity score under imperfect references and favour a correspondence-aware alternative (Sec. 8).
3. Prefer multi-annotator or bias-audited references for any benchmark used to select a model for clinical use, and document annotation provenance as first-class metadata, since it bounds the bias term μ no model engineering can undo.

Chapter 8

Directions

We separate what we can already recommend from what we can only propose.

A correspondence-aware topological metric (concrete). The Betti-0 pathology of Sec. 5.2.2 has a clear cause – $|\Delta N|$ compares counts, not components – and a clear remedy: measure topology in a way that pairs structures before scoring them. Two options are practical. (i) *clDice* is already correspondence-aware (it tests whether one skeleton lies inside the other volume) and, in our data, behaved correctly where $|\Delta N|$ did not; we recommend it as the default connectivity instrument under imperfect references. (ii) For a signed topological error, a *persistent-homology* Betti-number computed after matching components by spatial overlap (e.g. a Hungarian assignment on the connected-component graphs) would report unmatched births/deaths rather than a bare count difference, and would not collapse for two mutually fragmented masks. Re-running our Q1 analysis with such a metric is the immediate next step and requires no new data.

The learned-bias experiment (out of scope, left to future work). Whether a model trained on a fixed, systematically biased reference *internalises* that bias is a sharp question our framework poses but does not answer here: the two fixed-bias models did not complete training in time for this submission (Sec. 6). A cheap probe – fine-tuning the clean-trained model on a frozen biased reference for a short budget and checking whether ΔV drifts in the direction of the bias – would cost a fraction of the pre-registered design and could motivate a properly powered follow-up, but does not itself belong in this paper.

A distance-based proxy for the reversal mechanism. Sec. 5.4 correlates the *volume* omitted from the reference with M_0's HD95 penalty (ρ=0.39) as a first, coarse test of the mechanism; a distance metric should correlate with a tighter, more linear proxy – the maximum distance from an omitted component to the rest of the reference tree, which is what HD95 actually integrates and volume only approximates. We flag this as the natural next refinement, not yet computed.

Grounding the operator in real annotations. Our strongest structural assumption is that a calibrated synthetic operator stands in for real annotator disagreement (Sec. 9). The decisive validation is to acquire a second, independent tracing of a PARSE subset, measure the true inter-observer ΔV, clDice and NSD, and check that a setting of θ reproduces them – not just their central value but their spatial structure. This would convert every "within-IAA" statement in this paper from calibrated-to-literature to calibrated-to-data.

Chapter 9

Limitations

We list the boundaries of the study so that its claims are read at the right scope.

- **Simulated, not observed, noise.** $\mathcal{N}$ is calibrated to an inter-observer agreement window but is a model of annotation error, not a sample of it. Real annotator populations may combine families and exhibit spatially structured bias our single-family design does not reproduce. Composed noises are deferred by construction (Sec. 4).
- **One anatomy, one dataset, one architecture.** Results are on PARSE pulmonary arteries with nnU-Net at a fixed configuration. The operator and protocol transfer; operating points and effect sizes must be re-calibrated elsewhere.
- **Sample size and power.** With $n{=}85$ OOF cases the paired tests have limited power for small effects; this is why non-significant contrasts are reported with an MDD rather than as null. The $\mu{=}0$ regime (Q2) is expected to sit near that resolution limit.
- **The surface/topology coupling is a confound by nature.** On thin vessels we cannot fully isolate a pure surface effect from a topological one; we treat this as a real property to be reported jointly, not a nuisance to be removed, but it does bound single-metric interpretation.
- **Learned bias (Q3) out of scope.** The fixed-bias models (M_3, M_4) that would test whether a network learns a systematic annotation bias did not complete training in time for this submission (M_3: one fold, stopped at epoch 95/500, no recoverable checkpoint; M_4: training never started). We leave this question, and its fixed-realisation design (Table 4.2), to future work.
- **Metric set.** We study four metrics chosen for the pairing argument; other instruments (e.g. boundary IoU, centreline-length ratios) may resolve regimes ours cannot.

Chapter 10

Conclusion

We framed annotation noise as a formal, calibrated operator with a single bias axis, turned three intuitive questions into pre-registered, instrument-matched tests, and surfaced metrological limits – realistic boundary error is sub-clinical for standard metrics, and surface and topology are inseparable on thin vessels. The numbers bear the argument out: on 85 out-of-fold cases the artificial evaluation penalty (Q1) is large and unambiguous (ΔclDice$=-0.10$, $d=-3.06$), while the comfortable assumption that unbiased noise "augments" (Q2) did not hold – it cost a little and gained nothing on this anatomy. Whether a systematic bias is itself learnable by the network (Q3) is left to future work. Beyond any single number, the work is a plea for measurement humility in segmentation: a model is only as trustworthy as the reference and the metric we judge it by, and both must be reported, matched, and doubted with the same rigour we spend on the model.

Appendix A

Formal metric definitions

Let $P, G \subseteq \Omega$ be predicted and reference foreground sets on a grid with spacing s. Write ∂X for the surface (voxels of X with a background 6-neighbour), $\mathrm{Sk}(X)$ for the morphological skeleton, and $N(X)$ for the number of 26-connected components.

clDice (centreline Dice, ↑). With topological precision and sensitivity

$$T_{\text{prec}} = \frac{|\mathrm{Sk}(P) \cap G|}{|\mathrm{Sk}(P)|}, \qquad T_{\text{sens}} = \frac{|\mathrm{Sk}(G) \cap P|}{|\mathrm{Sk}(G)|},$$

clDice $= 2\,T_{\text{prec}} T_{\text{sens}}/(T_{\text{prec}} + T_{\text{sens}})$. A break in a tube is penalised even if it removes few voxels; a boundary shift that leaves the centreline intact is not – the source of clDice's blindness to drift.

NSD@τ (Normalised Surface Dice, ↑). With $d(x, Y) = \min_{y \in Y} \|x - y\|_s$,

$$\mathrm{NSD}_\tau = \frac{|\{x \in \partial P : d(x, \partial G) \le \tau\}| + |\{y \in \partial G : d(y, \partial P) \le \tau\}|}{|\partial P| + |\partial G|}.$$

We use $\tau = 0.5\,\mathrm{mm}$ (sub-voxel, sensitive to drift) and $\tau = 2\,\mathrm{mm}$ (clinical tolerance). NSD is *unsigned*: it cannot distinguish an inward from an outward displacement.

HD95 (↓, mm). The 95th percentile of $\{d(x, \partial G) : x \in \partial P\} \cup \{d(y, \partial P) : y \in \partial G\}$. Robust to 5% outliers; saturates below one voxel.

Betti$_0$ error (↓). $|N(P) - N(G)|$, counting fragmentations and fusions. Computed in 26-connectivity, for which $N(\mathrm{GT}^*){=}1$ on the connected arterial tree (6-connectivity spuriously reports ~ 99).

Signed volume change (ΔV). $\Delta V = (|P| - |G|)/|G|$. The only signed instrument in our set: $\Delta V < 0$ under-segmentation, $\Delta V > 0$ over-segmentation; monotone in the bias μ, hence the matched direction gauge (Table 4.1).

Appendix B

Degradation algorithms

Both operators are deterministic given $(\varphi, r, p, \mu, \text{seed})$.

`distal_omission`$(G; r, p)$**.** (i) Opening $\gamma_r = \delta_r \circ \varepsilon_r$ with a ball of radius r; (ii) removable set $R = G \setminus \gamma_r(G)$ (the structures thinner than r); (iii) partition R into connected components $\{C_c\}$; (iv) draw $b_c \sim \text{Bernoulli}(p)$ and set $\text{GT}^- = G \setminus \bigcup_{c:b_c=1} C_c$; (v) if $p > 0$ and no component was drawn, force the removal of one (the "at least one" guarantee). Biased, foreground-reducing, topology-altering.

`boundary_drift`$(G; r, p, \mu)$**.** (i) Signed distance $\phi = \text{SDF}(\partial G)$, $\phi < 0$ inside; (ii) a zero-mean Gaussian field smoothed to correlation length ℓ gives the fluctuation ξ, $\mathbb{E}[\xi] = 0$; (iii) a coverage gate g_p selects the top-p fraction of surface voxels; (iv) displacement $d(x) = a(\mu + \xi(x))g_p(x)$ with amplitude $a = r\bar{s}$; (v) $\text{GT}^- = \{x : \phi(x) - d(x) \leq 0\}$. By design the systematic part $a\mu g_p$ is a stationary target (in principle learnable) and the zero-mean part $a\xi g_p$ varies per realisation (augmentation-like); whether these design intents hold empirically is the object of Q2/Q3.

Appendix C

Extended calibration grid

Table C.1 reports the full grid measured on PARSE (`results/noise_calibration.csv`); the operating points of the main text are the rows landing in the target inter-observer window.

Table C.1: Noise-calibration grid on PARSE (n=8 cases). Each family is calibrated on its matched metric (omission on clDice, drift on NSD@0.5), both landing in $[0.85, 0.90]$ at the chosen point. `distal_truncation` is measured but discarded (redundant with omission; its very high clDice reflects a mismatched instrument – clDice is blind to edge filing – not harmlessness).

Family	(r, p, μ)	clDice	NSD@0.5	HD95	Betti_0
`distal_omission`	$(1, 0.5, 0)$	0.905	0.953	0.00	64.0
`distal_omission`	$(2, 0.3, 0)$	**0.872**	0.918	0.30	15.4
`distal_omission`	$(3, 0.3, 0)$	0.840	0.886	3.30	2.3
`distal_truncation`	$(2, 0.3, 0)$	0.991	0.985	0.00	0.0
`distal_truncation`	$(3, 0.5, 0)$	0.976	0.986	0.00	0.0
`boundary_drift`	$(1, 0.5, -1.0)$	0.936	0.840	0.58	142.9
`boundary_drift`	$(1, 0.5, -0.5)$	0.962	**0.885**	0.08	89.3
`boundary_drift`	$(1, 0.5, \ 0.0)$	0.978	0.898	0.00	50.3
`boundary_drift`	$(1, 0.5, +0.5)$	0.986	**0.869**	0.33	19.6
`boundary_drift`	$(1, 0.5, +1.0)$	0.985	0.820	0.66	6.9

Appendix D

Statistical protocol

Evaluation is *out-of-fold*: each of five nnU-Net folds predicts the ~ 17 validation cases it never trained on; their union covers all 85 cases without leakage, and n grows monotonically as folds finish (the *anytime* property). For each pre-registered contrast we compute per-case paired differences and test H_0 : median $= 0$ with the Wilcoxon signed-rank test (no normality assumption); effect size is the paired Cohen's $d = \overline{\Delta}/s_\Delta$. Across the family of pre-specified tests we control the false discovery rate at q=0.05 (Benjamini–Hochberg). When a contrast is non-significant we report the minimum detectable difference (MDD) at 80% power given the observed paired variance, and interpret any model gap below the MDD as *undecided* rather than null. Pre-specified outcome deltas additionally carry a bootstrap 95% confidence interval. Everything is regenerated from the master table by `report.py`; results are watermarked PRELIMINARY until all folds are in.

Appendix E

Full evaluation-penalty matrix

Table E.1 extends Q1 to every model and every degraded reference: each cell is the paired change (degraded reference $-$ GT*) on the matched instruments, with FDR-corrected significance. *Every* contrast is significant at $p_{\text{FDR}} < 0.001$, and the pattern is consistent across the three models – confirming that the evaluation penalty is driven by the reference, not by which training regime produced the predictions. The signed ΔV column changes sign between the receded (μ<0, $+$) and advanced (μ>0, $-$) drift references, the direction information that the unsigned surface metrics discard.

Table E.1: Evaluation penalty (degraded reference minus GT*), OOF n=85, paired Wilcoxon with Benjamini–Hochberg FDR ($^{*}p$<0.05, $^{**}p$<0.01, $^{***}p$<0.001). clDice and NSD@0.5 negative = the same network scored worse; ΔV signed.

Model	Reference	ΔclDice	ΔNSD@0.5	$\Delta(\Delta V)$
M_0	omission	-0.100^{***}	-0.055^{***}	$+0.076^{***}$
M_0	drift$-$	-0.029^{***}	-0.054^{***}	$+0.045^{***}$
M_0	drift$+$	-0.009^{***}	-0.061^{***}	-0.067^{***}
M_1	omission	-0.086^{***}	-0.043^{***}	$+0.072^{***}$
M_1	drift$-$	-0.020^{***}	-0.048^{***}	$+0.043^{***}$
M_1	drift$+$	-0.009^{***}	-0.065^{***}	-0.063^{***}
M_2	omission	-0.086^{***}	-0.042^{***}	$+0.071^{***}$
M_2	drift$-$	-0.021^{***}	-0.047^{***}	$+0.042^{***}$
M_2	drift$+$	-0.009^{***}	-0.066^{***}	-0.063^{***}

The counterpoint: model effects on a fixed reference. Table E.2 reports the same paired tests *between models* on the clean GT*. The clean baseline M_0 is significantly but slightly ahead of the two noisy-trained models (ΔclDice ≈ -0.02), while the two noisy regimes M_1 and M_2 are statistically *indistinguishable* on every metric, and the topological error (Betti$_0$) separates no pair of models. Placed beside Table E.1, the contrast is stark: the largest between-*model* effect on a fixed reference (~ 0.02 clDice) is five times smaller than the smallest Q1 reference penalty for M_0 and twenty times smaller than the largest – the quantitative core of the paper.

Table E.2: Inter-model contrasts on the clean reference GT*, OOF n=85 (same test and FDR as Table E.1). Differences are small; M_1/M_2 do not separate.

Pair	ΔclDice	ΔNSD@0.5	ΔBetti$_0$
M_0/M_1	-0.018 ***	-0.022 ***	$+2.0$ n.s.
M_0/M_2	-0.019 ***	-0.023 ***	$+1.4$ n.s.
M_1/M_2	-0.001 n.s.	-0.001 n.s.	-0.7 n.s.

Appendix F

Reproducibility

Every figure and number regenerates from the raw nnU-Net predictions with a fixed seed. The pipeline is: (i) `generate_degraded_dataset.py -seed 42` materialises the deterministic evaluation references GT^{-} for each scenario; (ii) `collect_metrics.py` (or its parallel variant, which crops each prediction/reference pair to their common bounding box – an exact optimisation verified to reproduce every metric to machine precision) writes the long-format master table `results/metrics.csv`, one row per $(\text{model}, \text{fold}, \text{scenario}, \text{case})$ carrying its provenance (epochs, checkpoint time, git commit, nnU-Net version); (iii) `report.py` emits the paired-Wilcoxon tables (`metrics_stats_oof.csv`), the pre-specified outcomes with bootstrap CIs (`outcomes_oof.csv`), the summary figures, and a `STATUS.md` dashboard watermarked by fold coverage. Evaluation is out-of-fold, so no case is scored by a model that trained on it; the master table grows monotonically as folds complete, and re-running any stage only computes the missing keys (idempotent upsert). All results in this paper are tier-A (M_0, M_1, M_2, n=85, all five folds); the tier-B fixed-bias models (M_3, M_4) did not complete training in time for this submission and no tier-B numbers are reported (Sec. 6).

Appendix G

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